

CONTACT

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CAREER**Postdoctoral Research Fellow**

Aug 2025-now

Department of Physics and Astronomy, University of Exeter | Exeter, UK

*Maternity leave (9 months)***Postdoctoral Research Fellow**

Apr 2022-Jul 2025

Department of Physics and Astronomy, University of Exeter | Exeter, UK

Postdoctoral Research Fellow

Sep 2019-Mar 2022

Department of Physics and Astronomy, University of Exeter | Exeter, UK

EDUCATION

PhD in Environmental Sciences

2015-2020

School of Environmental Sciences, University of East Anglia | Norwich, UK

Supervisors: Prof. Claire Reeves, Dr Paul Griffiths, Dr Marcus Köhler, Dr Mike Newland

Thesis: Impacts of C₁-C₃ alkyl nitrates on tropospheric ozone chemistry**MSc in Climate Change with Distinction**

2014-2015

School of Environmental Sciences, University of East Anglia | Norwich, UK

Supervisor: Prof. Claire Reeves

Thesis: Investigation of the relationship between tropospheric ozone production efficiency and carbon bond emissions

Specialist Diploma in Meteorology

2009-2014

Faculty of Geography, Lomonosov Moscow State University | Moscow, Russia

Supervisor: Prof. Alexander V. Kislov

Thesis: Climatically-induced variations of the Caspian Sea level over the last Millennium

PUBLICATIONS

17. Mukherjee S., Sing D. K., Fu G., Stevenson K. B. et al. (incl. **Zamyatina M.**) (submitted). [Cloudy mornings and clear evenings on a giant extrasolar world.](#)
16. Fu G., Mukherjee S., Stevenson K. B., Sing D. K. et al. (incl. **Zamyatina M.**) (2025). [Overcast mornings and clear evenings in hot Jupiter exoplanet atmospheres.](#) ApJL.
15. Mak M. T., Sergeev D. E., Mayne N. J., **Zamyatina M.**, Steinrueck M. E. et al. (2025). [The impact of different haze types on the atmosphere and observations of hot Jupiters: 3D simulations of HD 189733b, HD 209458b and WASP-39b.](#) MNRAS.
14. Ahrer, E-M., Gandhi, S., Alderson, L., Kirk, J. et al. (incl. **Zamyatina, M.**) (2025). [Tracing the formation and migration history: molecular signatures in the atmosphere of misaligned hot Jupiter WASP-94Ab using JWST NIRSpec/G395H.](#) MNRAS.
13. Meech, A., Claringbold, A. B., Ahrer, E-M., Kirk, J. et al. (incl. **Zamyatina, M.**) (2025). [BOWIE-ALIGN: Sub-stellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy.](#) MNRAS.
12. Kirk, J., Ahrer, E-M., Claringbold, A. B., **Zamyatina, M.**, Fisher C. et al. (2025). [BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b.](#) MNRAS.
11. Penzlin, A. B. T., Booth, R. A., Kirk, J., Owen, J. E. et al. (incl. **Zamyatina, M.**) (2024). [BOWIE-ALIGN: how formation and migration histories of giant planets impact atmospheric compositions.](#) MNRAS.
10. Kirk, J., Ahrer, E-M., Penzlin, A. B. T., Owen, J. E. et al. (incl. **Zamyatina, M.**) (2024). [BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history.](#) RAS Techniques and Instruments.
9. Espinoza, N., Steinrueck, M., Kirk, J., MacDonald, R. J. et al. (incl. **Zamyatina, M.**) (2024). [Inhomogeneous terminators on the exoplanet WASP-39 b.](#) Nature.
8. Christie, D. A., Mayne, N. J., **Zamyatina, M.**, Baskett, H., Evans-Soma, T. M., et al. (2024). [Longitudinal filtering, sponge layers, and equatorial jet formation in a general circulation model of gaseous exoplanets.](#) MNRAS.
7. **Zamyatina, M.**, Christie, D. A., Hébrard, E., Mayne, N. J., Radica, M. et al. (2024). [Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example.](#) MNRAS.
6. Taylor, J., Radica, M., Welbanks, L., MacDonald, R. J. et al. (incl. **Zamyatina, M.**) (2023). [Awesome SOSS: atmospheric characterisation of WASP-96b using the JWST early release observations.](#) MNRAS.
5. Radica, M., Welbanks, L., Espinoza, N., Taylor, J. et al. (incl. **Zamyatina, M.**) (2023).

[Awesome SOSS: transmission spectroscopy of WASP-96b with NIRISS/SOSS](#). MNRAS.

4. **Zamyatina, M.**, Hébrard, E., Drummond, B., Mayne, N. J., Manners, J. et al. (2023). [Observability of signatures of transport-induced chemistry in clear atmospheres of hot gas giant exoplanets](#). MNRAS.
3. Christie, D. A., Lee, E. K. H., Innes, H., Noti, P. A. et al. (incl. **Zamyatina, M.**) (2022). [CAMEMBERT: A Mini-Neptunes GCM Intercomparison, Protocol Version 1.0. A CUISINES Model Intercomparison Project](#). Planet. Sci. J.
2. Ridgway, R. J., **Zamyatina, M.**, Mayne, N. J., Manners, J., Lambert, F. H. et al. (2023). [3D modelling of the impact of stellar activity on tidally locked terrestrial exoplanets: atmospheric composition and habitability](#). MNRAS.
1. Braam, M., Palmer, P. I., Decin, L., Ridgway, R. J., **Zamyatina, M.** et al. (2022). [Lightning-induced chemistry on tidally-locked Earth-like exoplanets](#). MNRAS.

INVITED TALKS

- Jul 2025 [Atmospheric chemistry of solar and extrasolar gas giants](#) (review)
[Exoclimes VII conference](#) | Montreal, Canada
- Mar 2024 Overview of the Met Office Unified Model configuration for hot Jupiter atmospheres
International Space Science Institute (ISSI) workshop | Bern, Switzerland
- Feb 2024 Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere
University of Bristol (astronomy seminar) | Bristol, UK
- Feb 2023 Atmospheric dynamics and chemistry on exoplanets
University of Queensland (astronomy seminar) | Brisbane, Australia
University of Southern Queensland (exoplanet seminar) | Brisbane, Australia
University of New South Wales (astronomy seminar) | Sydney, Australia
- Nov 2022 Observability of signatures of wind-driven chemistry in atmospheres of hot gas giants
Ludwig Maximilian University (exoplanet group seminar) | Munich, Germany
[Celebrating JWST's first six months of exoplanet data](#) workshop | Ringberg castle, Germany
- Oct 2022 Modelling chemistry of hot Jupiter atmospheres with the Met Office Unified Model
Met Office | Exeter, UK
- Feb 2022 Transport-induced quenching shapes transmission spectra of warm and hot Jupiters
University of Warwick (astronomy seminar) | virtual

CONTRIBUTED TALKS

- Sep 2023 Metallicity masquerade: how to use quenching to distinguish between different planet metallicities
University of Bristol (BOWIE meeting) | Bristol, UK
- June 2021 Overview of the Met Office Unified Model adapted to simulate exoplanetary atmospheres
Ariel consortium meeting | virtual
- Apr, Sep 2021 [3D simulations of warm and hot Jupiter atmospheres: the role of 3D mixing in shaping CH₄-to-CO conversion pathways](#)
EPSC conference | virtual
UKEXOM conference | virtual
University of Exeter (astronomy seminar) | Exeter, UK
- Mar, Apr, Jun 2019 [Impact of C₁-C₃ alkyl nitrate chemistry on tropospheric ozone: box and global model perspectives](#)
University of Exeter (XCS seminar) | Exeter, UK
EGU conference | Vienna, Austria
University of East Anglia (AMB seminar) | Norwich, UK
- Apr 2017 Adding new chemistry into UM-UKCA
Cambridge-EnvEast doctoral alliance symposium | Cambridge, UK
- Sep 2012 Assessment of climatological potential of transboundary air pollution transport in Eastern Siberia and the Russian Far East
[Air quality management at urban, regional and global scales 4th international symposium/IUAPPA regional conference](#) | Istanbul, Turkey

POSTERS

- Jul 2025 [CH₄-CO conversion pathways in hot Jupiter atmospheres](#)
[Exoclimes VII conference](#) | Montreal, Canada
- Apr, Jun 2024 [Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere](#)
UKEXOM conference | Birmingham, UK
Exoplanets 5 conference | Leiden, Netherlands
- Sep 2022 [Applying known chemical kinetics data to model atmospheres of extrasolar planets](#)
iCACGP-IGAC conference | Manchester, UK
- Sep 2021 [Local and global impacts of C₁-C₃ alkyl nitrate photochemistry and emissions on tropospheric ozone](#)
IGAC conference | virtual
- Sep 2018 [Impact of alkyl nitrate chemistry on tropospheric ozone](#)
iCACGP-IGAC conference | Takamatsu, Japan

Mar, Apr 2018	Impact of C ₁ -C ₅ alkyl nitrate chemistry on tropospheric ozone - a box modelling study Cambridge-EnvEast doctoral alliance symposium Cambridge, UK EGU conference Vienna, Austria		
AWARDS	2023	Above & Beyond Award	
	2022	EPSRC vacation internship (for 3 interns)	12893.55£
	2022	Jackson-Grime-Davies (JGD) research internship (for 1 intern)	2428.71£
	2021	IGAC Early Career Scientist poster prize & travel grant	1227.70£
	2015-2019	Lord Zuckerman studentship	112269.50£
	2014-2015	Simon Wharmby postgraduate scholarship	3000.00£
	2012	World Meteorological Organization travel grant	1154.10£
AWARDED OBSERVING TIME	Feb 2024	JWST's exoplanet grand tour spectroscopic survey	
		▪ Co-I HST GO-17612 (PI: David Sing)	24 orbits
		▪ Co-I JWST GO-5924 (PI: David Sing)	125.70 hours
	Feb 2024	Starspots, hazes, and disequilibrium chemistry: a deep dive into the atmosphere of HAT-P-18b	
		▪ Co-I JWST GO-5844 (PI: Michael Radica)	16.40 hours
	May 2023	Putting it all together: Dynamics and chemistry probed through transmission spectroscopy of a cloud-free exoplanet	
		▪ Co-I JWST GO-4082 (PI: Michael Radica, Co-PI: Jake Taylor)	6.69 hours
	May 2023	Hot Jupiter atmospheric forecast: are mornings cloudier than evenings in other worlds?	
		▪ Co-I JWST GO-3969 (PI: Nestor Espinoza, Co-PI: Diana Powell)	61.53 hours
	May 2023	Does atmospheric composition actually trace formation? Observing aligned vs misaligned hot Jupiters as a testbed	
		▪ Co-I JWST GO-3838 (PI: James Kirk Co-PI: Eva-Maria Ahrer)	49.21 hours
	May 2023	Testing the C/O ratio prediction for hot Jupiters from disk-free migration	
		▪ Co-I JWST GO-3154 (PI: Eva-Maria Ahrer)	10.36 hours
SUPERVISION	Primary supervisor and co-supervisor. Students who went on to do a PhD are marked with *.		
		PhD supervision (2)	
	Sep 2024-now	Harry Baskett Thesis: TBD Co-supervisors: Dr. E. Hebrard, Prof. N. J. Mayne	
	Nov 2020-May 2023	Robert J. Ridgway Thesis: Simulating the impact of stellar flares on the climate and habitability of terrestrial Earth-like exoplanets Co-supervisors: Prof. N. J. Mayne, Prof. F. H. Lambert, Dr. J. Manners	
		Undergraduate and summer internship supervision (4)	
	Jun-Aug 2022	EPSRC-funded: Harry Baskett*, Ben Moore*, James McDermott*; JGD-funded: Graig Lils Project: 3D modelling of hot Saturn atmospheric chemistry	
TEACHING	Jul 2023	Module leader Module: No place like home: placing Earth in its geological and astronomical contexts International sustainability summer school University of Exeter, Exeter, UK	
	Jul 2022, Jul 2023	Lecturer Module: No place like home: placing Earth in its geological and astronomical contexts International sustainability summer school University of Exeter, Exeter, UK	
	Sep 2021-Feb 2022	Associate Tutor Modules: Experimental science, Frontiers in science University of Exeter Exeter, UK	
	Jan 2018	Instructor Module: Introduction to Python in Environmental Sciences University of East Anglia Norwich, UK	
	2015-2018	Associate Tutor Modules: Numerical skills for scientists, Atmospheric chemistry and global change, Atmospheric composition (measurements and modelling), Atmosphere & oceans I University of East Anglia Norwich, UK	
ACADEMIC COMMUNITY	Organisation of scientific meetings		
	26-30 Jun 2023	Exoclimes VI conference (LOC member, session chair) University of Exeter Exeter, UK	~200 attendees

	22-24 Jun 2023	ExoSLAM school (LOC member) University of Exeter Exeter, UK	~50 attendees
	5-6 Dec 2022	BOWIE meeting (co-organiser) University of Exeter Exeter, UK	17 attendees
	Sep 2017-Jun 2018	Atmospheric and Marine Biogeochemistry (AMB) seminars (co-organiser) University of East Anglia Norwich, UK	~20 attendees
		Reviewing Journals: The Astrophysical Journal, Monthly Notices of the Royal Astronomical Society Proposals: JWST Cycle 3 TAC external expert, JWST Cycle 4 TAC panelist	
OUTREACH	Jul 2024	Joint press release about Espinoza et al. (2024) paper: Research confirms that distant world's eternal sunrise and sunset are not alike	
	Sep 2023	Expert scientist at the Climate Exhibition (part of the British Science Festival)	
	Nov 2015-Jun 2019	Maintainer of @AtmosChemUEA Twitter account	
VOCATIONAL EXPERIENCE	Aug-Sep 2013	Weather Forecaster Sheremetyevo International Airport Moscow, Russia	
	Jun-Jul 2013	Technician Department of Actinometry, Meteorological Observatory Lomonosov Moscow State University Moscow, Russia	
VOCATIONAL TRAINING	Sep 2023	Belbin training	
	Mar 2023	Leadership training	
	Dec 2022	Interview training	
	Sep 2021	Learning and Teaching in Higher Education (LTHE) Unit 1	
	Mar 2020	JWST proposal planning workshop	
	Apr 2016	NAME workshop	
	Jan 2016	Introduction to UKCA	
	Dec 2015	Introduction to Unified Model	
	Nov 2015	Introduction to Atmospheric Science	
	2015-2019	EnvEast Doctoral Training Programme	